
SMART HR OPERATIONS AND WORKFORCE MANAGEMENT SYSTEM

Software Design and Architecture

Deliverable 2

Group Number # 16

Ms. Laiba Imran

Employee Management	Leave & Attendance	Performance & Compliance
Sohaib(24i-3108) UC-01 to UC-05	Shaiman(24i-3074) UC-06 to UC-10	Hasaam(24i-3107) UC-11 to UC-15

Table of Contents

1. Introduction.....	4
2. Brief Use Cases (UC-01 to UC-15)	5
2.1 Employee Management Module (UC-01 – UC-05).....	5
UC-01 — Register New Employee.....	5
UC-02 — Update Employee Record	5
UC-03 — Assign Employee to Department	6
UC-04 — Initiate Employee Offboarding	6
UC-05 — View Employee Profile.....	6
2.2 Leave and Attendance Management Module (UC-06 – UC-10).....	7
UC-06 — Submit Leave Request	7
UC-07 — Approve or Reject Leave Request	7
UC-08 — Record Daily Attendance	7
UC-09 — Request Attendance Correction	8
UC-10 — View Leave Balance and Attendance Summary.....	8
2.3 Performance and Compliance Management Module (UC-11 – UC-15)	8
UC-11 — Initiate Performance Evaluation Cycle	9
UC-12 — Submit Employee Performance Evaluation	9
UC-13 — Monitor Employee Probation Status	9
UC-14 — Generate Compliance Report	10
UC-15 — Generate HR Analytics Report	10
3. Fully Dressed Use Cases (UC-01 to UC-15).....	11
3.1 Employee Management Module (UC-01 – UC-05).....	11
UC-01 Register New Employee.....	11
UC-02 Update Employee Record.....	14
UC-03 Assign Employee to Department.....	16
UC-04 Initiate Employee Offboarding	18
UC-05 View Employee Profile	20
3.2 Leave and Attendance Management Module (UC-06 – UC-10).....	23
UC-06 Submit Leave Request.....	23
UC-07 Approve or Reject Leave Request.....	25
UC-08 Record Daily Attendance	27

UC-09 Request Attendance Correction.....	29
UC-10 View Leave Balance and Attendance Summary	31
3.3 Performance and Compliance Management Module (UC-11 – UC-15)	34
UC-11 Initiate Performance Evaluation Cycle	34
UC-12 Submit Employee Performance Evaluation	36
UC-13 Monitor Employee Probation Status	38
UC-14 Generate Compliance Report	40
UC-15 Generate HR Analytics Report.....	43
4. Domain Model Summary.....	46
4.1 Core Entities and Key Attributes	46
4.2 Key Relationships	47
4.3 Domain Model	49
5. Use Case Diagram.....	50

1. Introduction

The document presents a use case specification for the Smart HR Operations and Workforce Management System, which functions as an enterprise platform that streamlines and automates fundamental human resource activities needed by organizations. The system digitizes core human resources functions which include managing employee life cycles and tracking employee leave and attendance and conducting performance assessments and creating compliance reports. The system achieves its operational goals through its design which combines all organizational functions into one centralized system that enhances both operational performance and organizational transparency and policy enforcement.

The document presents use cases which demonstrate how various system users achieve their business objectives through system operations. The system identifies three main user groups which are Admin users and HR staff and regular Employees who access system functions based on their designated user roles.

The document contains four main sections that define its structure. The first section introduces the system and the purpose of this specification. The second section presents all use cases in brief format, providing a concise overview of the actors and interactions involved. The third section contains the fully dressed use cases, which describe each use case in detail, including stakeholders, preconditions, success scenarios, alternative flows, and system

2. Brief Use Cases (UC-01 to UC-15)

The following table-based summaries present all fifteen use cases in brief format. The brief format captures the use case identity, scope, actor, type, and a concise description sufficient for stakeholder review and initial backlog estimation.

2.1 Employee Management Module (UC-01 – UC-05)

UC-01 — Register New Employee

Use Case ID	UC-01
Use Case Name	Register New Employee
Actors	HR
Type	Primary
Description	The HR Officer registers a new employee by entering personal details, department, designation, and employment type. The system validates all fields for completeness and uniqueness, generates a unique Employee ID, persists the record, and dispatches an automated welcome notification to the new employee.

UC-02 — Update Employee Record

Use Case ID	UC-02
Use Case Name	Update Employee Record
Actors	HR
Type	Primary
Description	The HR Officer locates an existing employee record and modifies one or more fields such as contact details, designation, or salary grade. The system validates the changes, writes a timestamped audit log capturing old and new values, and saves the updated record.

UC-03 — Assign Employee to Department

Use Case ID	UC-03
Use Case Name	Assign Employee to Department
Actors	HR(Primary), Admin
Type	Primary
Description	The Admin or HR Officer assigns or transfers an employee to a department, specifying an effective date. The system validates departmental capacity, updates the organizational hierarchy, records the assignment history, and notifies the relevant managers.

UC-04 — Initiate Employee Offboarding

Use Case ID	UC-04
Use Case Name	Initiate Employee Offboarding
Actors	HR
Type	Primary
Description	The HR Officer starts the offboarding workflow for a departing employee by specifying the separation type, effective date, and exit reason. The system creates an offboarding checklist, schedules access revocation, triggers the final settlement workflow, and updates the employee status.

UC-05 — View Employee Profile

Use Case ID	UC-05
Use Case Name	View Employee Profile
Actors	HR(Primary), Employee
Type	Primary
Description	An authenticated user accesses a consolidated employee profile showing personal details, employment history, department, leave balances, attendance summary, and performance ratings. The system enforces role-based field masking to restrict sensitive information.

2.2 Leave and Attendance Management Module (UC-06 – UC-10)

UC-06 — Submit Leave Request

Use Case ID	UC-06
Use Case Name	Submit Leave Request
Actors	Employee
Type	Primary
Description	The Employee submits a formal leave request by selecting leave type and dates. The system validates the available balance and policy calendar, records the request as Pending Approval, blocks the dates in the attendance calendar, and notifies the assigned HR Officer.

UC-07 — Approve or Reject Leave Request

Use Case ID	UC-07
Use Case Name	Approve or Reject Leave Request
Actors	HR
Type	Primary
Description	The HR Officer reviews a pending leave request and approves or rejects it with a mandatory comment. On approval the system deducts the leave balance, updates the attendance calendar, and notifies the employee. On rejection the record is retained with the reason for audit.

UC-08 — Record Daily Attendance

Use Case ID	UC-08
Use Case Name	Record Daily Attendance
Actors	Employee
Type	Primary

Description	The system records check-in and check-out timestamps for each employee via biometric device, web portal, or mobile app. It computes work hours, flags late arrivals and early departures, and stores a structured attendance record linked to payroll.
--------------------	--

UC-09 — Request Attendance Correction

Use Case ID	UC-09
Use Case Name	Request Attendance Correction
Actors	Employee
Type	Primary
Description	If an attendance record contains an error, the Employee submits a correction request specifying the affected date, the incorrect value, the corrected value, and a supporting reason. HR reviews the request and approved corrections update the record with an audit annotation.

UC-10 — View Leave Balance and Attendance Summary

Use Case ID	UC-10
Use Case Name	View Leave Balance and Attendance Summary
Actors	Employee, HR
Type	Primary
Description	Authorized users access a real-time dashboard showing remaining leave balances by type and monthly attendance KPIs. The system aggregates data from the leave and attendance modules and supports export to PDF or Excel.

2.3 Performance and Compliance Management Module (UC-11 – UC-15)

UC-11 — Initiate Performance Evaluation Cycle

Use Case ID	UC-11
Use Case Name	Initiate Performance Evaluation Cycle
Actors	Admin
Type	Primary
Description	The admin configures and launches a performance evaluation cycle by defining the period, scope, evaluators, and deadlines. The system distributes evaluation tasks, opens the evaluation module, and schedules automated reminder notifications to evaluators.

UC-12 — Submit Employee Performance Evaluation

Use Case ID	UC-12
Use Case Name	Submit Employee Performance Evaluation
Actors	HR
Type	Primary
Description	The HR (assigned evaluator) completes a structured performance evaluation form for an employee during an active evaluation cycle by scoring competencies and KPIs. The system validates completeness, stores the evaluation, calculates an aggregate score, and records it in the employee performance history.

UC-13 — Monitor Employee Probation Status

Use Case ID	UC-13
Use Case Name	Monitor Employee Probation Status
Actors	HR (Primary), Admin (Supporting)
Type	Primary
Description	The Admin or HR Officer monitors employees on probation, tracking evaluation milestones and decision dates. The system sends automated deadline alerts and allows HR to record confirmation, extension, or termination decisions with supporting documentation.

UC-14 — Generate Compliance Report

Use Case ID	UC-14
Use Case Name	Generate Compliance Report
Actors	Admin
Type	Primary
Description	The admin generates a formal compliance report for a specified period and compliance domain. The system aggregates policy adherence data, applies regulatory filters, formats the output in a time-stamped report, and archives it for a minimum of five years.

UC-15 — Generate HR Analytics Report

Use Case ID	UC-15
Use Case Name	Generate HR Analytics Report
Actors	Admin
Type	Primary
Description	The admin generates an analytics report covering workforce KPIs such as headcount, attrition, leave utilization, and performance benchmarks. The system queries all HR modules, applies statistical aggregation, renders interactive charts and data tables, and supports export and scheduled delivery.

3. Fully Dressed Use Cases (UC-01 to UC-15)

This section presents each use case in the fully dressed format. The fully dressed template captures the complete behavioral specification including all actor classifications (Primary, Supporting, and Offstage), stakeholder interests, preconditions, success guarantee, main success scenario, alternative and exception flows (minimum three extensions per use case), non-functional requirements, and open design issues.

3.1 Employee Management Module (UC-01 – UC-05)

UC-01 | Register New Employee

Use Case ID	UC-01
Use Case Name	Register New Employee
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	- System Administrator - IT Department	- New Employee - Payroll Module - Compliance Officer

Stakeholders and Interests

- HR Officer – wants to enter complete, accurate employee data quickly and without duplicates.
- Admin – wants all new employees captured before access or payroll activation.
- New Employee – wants their profile created correctly so benefits, access, and salary are processed from day one.

- IT Department – wants automatic user-account provisioning triggered upon registration.
- Compliance Officer – wants mandatory fields enforced and data captured per regulatory requirements.

Preconditions

- The HR Officer is authenticated and holds the HR Manager or HR Executive role.
- No existing record with the same National ID or email address exists in the system.
- The target department and designation are present in the system master data.
- The employment contract has been signed and scanned copies are available for upload.

Success Guarantee (Postconditions)

- A unique Employee ID is generated and permanently assigned to the record.
- The employee record is persisted in the database with all provided information.
- An automated welcome email is dispatched to the new employee.
- The employee is listed as Active and appears in the relevant department roster.
- A user account is provisioned with default role-based access rights.

Main Success Scenario

Step	Actor Action	System Response
1	HR Officer navigates to Register New Employee module.	System displays the structured employee registration form highlighting all mandatory fields.
2	HR Officer enters personal details: full name, national ID, date of birth, gender, contact number, and residential address.	System performs real-time duplicate check on national ID and email; immediately highlights any conflicts.
3	HR Officer selects department, designation, employment type, and joining date.	System validates selections against master data and auto-populates probation period based on employment type.
4	HR Officer uploads supporting documents (signed contract, ID scan).	System validates file format and size, attaches documents to the record.
5	HR Officer clicks Submit to finalize the registration.	System generates unique Employee ID, saves the record, triggers welcome email, provisions user account, and displays a success confirmation with the new Employee ID.

Extensions (Alternative Flows)

- 2a. Duplicate National ID detected: System displays an error banner and prevents submission until the conflict is resolved by HR or Admin.
- 2b. Email address already registered to another employee: System flags the duplicate email field with an inline warning and requires correction.
- 3a. Selected department has reached maximum headcount: System displays a capacity warning; HR may override with explicit Admin authorization and a logged justification.
- 4a. Uploaded document format is not supported (e.g., not PDF or JPEG): System displays accepted formats and prompts re-upload without clearing other fields.
- 4b. Document file size exceeds the 5 MB limit: System rejects the file with a size-limit message and suggests compression.
- 5a. System fails to send the welcome email due to SMTP unavailability: Registration is saved successfully and the email is queued for automatic retry; HR is informed of the queue status.

Special Requirements

- Employee ID must follow the organizational format EMP-YYYY-NNNN and be globally unique across all entities.
- All form submissions must complete within 3 seconds under normal system load.
- All data must be encrypted in transit using TLS 1.2 or higher and encrypted at rest.

Technology & Data Variations	Web interface supporting Chrome, Firefox, and Edge.
Frequency of Occurrence	Low to medium — typically 5 to 30 registrations per month.
Open Issues	Automatic salary structure assignment via payroll integration upon registration is pending finalization.

UC-02 | Update Employee Record

Use Case ID	UC-02
Use Case Name	Update Employee Record
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	- Admin (override for restricted fields) - System (audit logger)	- Payroll Module - Compliance Officer (audit reviewer) - Employee

Stakeholders and Interests

- HR Officer – wants to modify employee data efficiently with a clear audit trail.
- Admin – wants all updates logged and attributable to a specific user.
- Employee – wants personal information kept current for payroll and communications.
- Compliance Officer – wants a complete change history available for regulatory audits.
- Payroll Module – wants instant notification of salary or designation changes.

Preconditions

- The HR Officer is authenticated and authorized to modify employee records.
- The target employee record exists and has an Active or On Leave status.
- The field(s) to be updated are within the HR Officer's role-based permission scope.

Success Guarantee (Postconditions)

- The employee record reflects the updated information.
- A timestamped audit log entry is stored capturing the field name, old value, new value, and updater ID.
- Payroll and reporting modules are notified if salary or designation has changed.
- The employee receives an automatic notification of profile updates that affect them.

Main Success Scenario

Step	Actor Action	System Response
1	HR Officer searches for the employee using Employee ID, name, or department filter.	System returns a list of matching records with summary details.
2	HR Officer selects the employee and clicks Edit Record.	System displays the editable profile form pre-populated with current data; restricted fields are visually disabled.
3	HR Officer modifies the required fields.	System validates each modified field in real time against defined format and constraint rules.
4	HR Officer provides a mandatory reason for the update and clicks Save Changes.	System saves the updated record, writes the audit log entry, notifies downstream modules, and shows a success confirmation.

Extensions (Alternative Flows)

- 3a. Invalid format for a modified field (e.g., malformed phone number): System highlights the field with an inline error and blocks form submission.
- 3b. HR Officer attempts to modify a salary-grade field without sufficient privileges: System displays Access Restricted for that field and logs the attempt.
- 3c. HR Officer attempts to update National ID: System requires secondary Admin approval; the change enters a Pending Approval state until Admin confirms.
- 4a. Concurrent edit detected (another user has modified the same record): System displays a conflict warning showing both versions and prompts HR to confirm which values to retain.
- 4b. Database write fails due to a connectivity issue: System shows an error message, preserves the entered data in session, and allows the HR Officer to retry without re-entering values.
- 4c. Downstream payroll notification fails: System saves the update successfully, queues the notification for retry, and alerts HR to verify payroll sync manually.

Special Requirements

- All updates must be logged with user ID, timestamp, and delta (before and after values) in an immutable audit table.
- Sensitive fields such as salary and national ID must require an additional confirmation step or secondary approval.

Technology & Data Variations	Web interface with inline validation; immutable audit log stored in a dedicated append-only database table.
Frequency of Occurrence	Medium — multiple updates expected daily across the organization.
Open Issues	Bulk update capability for mass changes during organizational restructuring is under design review.

UC-03 | Assign Employee to Department

Use Case ID	UC-03
Use Case Name	Assign Employee to Department
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	- System (hierarchy updater) - IT (access sync)	- Previous Department Manager - New Department Manager - Employee

Stakeholders and Interests

- Admin – wants a clean organizational structure with accurate reporting hierarchies.
- HR Officer – wants assignment history available for workforce planning and auditing.
- Employee – wants to be correctly assigned so access rights, line manager, and team membership are accurate.
- Department Managers – want real-time visibility of team headcount changes.
- IT Department – wants access-rights synchronization triggered automatically.

Preconditions

- The employee exists in the system with an Active employment status.
- The target department is registered in the organizational structure master data.

- The assigning user holds Org Management or HR Admin permissions.

Success Guarantee (Postconditions)

- The employee departmental association is updated in the system.
- The transfer is recorded in the employee organizational history with an effective date.
- Reporting line and access rights are updated to reflect the new department.
- Both the previous and new department managers are notified.

Main Success Scenario

Step	Actor Action	System Response
1	Admin navigates to Organizational Structure and selects the target department.	System displays department details including current headcount, reporting manager, and available slots.
2	Admin searches for and selects the employee to assign or transfer.	System displays the employee current department assignment and full employment summary.
3	Admin sets the effective date and optionally adds a remark.	System validates the effective date is not in the past and the department has capacity for the employee.
4	Admin confirms the assignment.	System updates the employee department, adjusts the reporting hierarchy, logs the change with the effective date, and dispatches notifications to both managers.

Extensions (Alternative Flows)

- 3a. Department has reached maximum headcount: System displays a capacity warning; Admin must document a justification and receive explicit override authorization.
- 3b. Effective date is backdated: System warns that a backdated assignment may affect historical records and requires Admin to submit an explicit acknowledgment.
- 3c. The employee is currently in an active offboarding workflow: System blocks the assignment and requires the offboarding to be cancelled first.
- 4a. Active Directory synchronization fails after assignment: System saves the organizational change and queues the AD sync for retry; IT is alerted via a system notification.
- 4b. The assigned employee has a dual reporting structure: System flags the conflict and requires Admin to resolve the primary reporting line before confirming.
- 4c. The new department requires a security clearance level the employee does not hold: System displays a clearance mismatch warning and prevents assignment until compliance is verified.

Special Requirements

- Assignment history must be immutable; corrections require a new superseding assignment record.
- Access-rights synchronization must complete within 5 minutes of a confirmed assignment.

Technology & Data Variations	Web interface with hierarchical org-chart visualization.
Frequency of Occurrence	Low to medium — typically during hiring cycles, restructuring events, or internal transfers.
Open Issues	Multi-department concurrent assignment for matrix organizations is scheduled for a future release.

UC-04 | Initiate Employee Offboarding

Use Case ID	UC-04
Use Case Name	Initiate Employee Offboarding
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	• Admin (system access revocation) • Finance • IT Security • Direct Manager	• Departing Employee • Payroll Module

Stakeholders and Interests

- HR Officer – wants a structured checklist ensuring no offboarding step is missed.
- Admin – wants system access revoked and assets recovered on or before the last working day.
- Finance – wants final settlement triggered with accurate deductions and entitlements computed.

- IT Security – wants all digital credentials and tokens invalidated immediately upon separation.
- Departing Employee – wants the process completed transparently and final dues settled correctly.

Preconditions

- The departing employee has an active record in the system.
- The HR Officer is authenticated and holds offboarding initiation privileges.
- A formal separation notice or approval exists (resignation letter, termination order, or contract expiry notice).

Success Guarantee (Postconditions)

- An offboarding workflow is created and assigned to all relevant stakeholders.
- System access revocation is scheduled or immediately executed for terminations.
- Final settlement computation is initiated in the payroll module.
- Employee status is updated to Offboarding – Pending Clearance.
- The employee record is archived upon full clearance completion.

Main Success Scenario

Step	Actor Action	System Response
1	HR Officer navigates to the employee profile and selects Initiate Offboarding.	System displays the offboarding form with separation type options and last working date selector.
2	HR Officer selects separation type (resignation, retirement, contract expiry, or termination) and enters the last working date.	System validates the last working date against notice period policies; displays any policy violations.
3	HR Officer assigns offboarding tasks to IT, Finance, and the employee (asset return, knowledge transfer, clearance signoff).	System generates the offboarding checklist, assigns tasks with individual due dates, and dispatches notification emails.
4	HR Officer submits the offboarding request.	System changes status to Offboarding – Pending Clearance, locks employment record editing, and triggers the final settlement workflow in the payroll module.

Extensions (Alternative Flows)

- 2a. Last working date violates the minimum notice period: System flags the violation and requires HR to acknowledge the exception or escalate for Admin approval.

- 2b. Separation type is Termination for Cause: System mandates upload of the formal termination letter before the workflow can proceed.
- 3a. An assigned IT or Finance stakeholder is unavailable or inactive: System allows HR to reassign the task to an alternate responsible party.
- 3b. Employee has outstanding financial obligations (advances, loans): System flags these in the checklist and requires Finance to confirm net settlement before archival.
- 4a. Payroll module is temporarily unavailable: Offboarding workflow is saved; payroll trigger is queued and retried automatically when connectivity is restored.
- 4b. Employee disputes the last working date or settlement amount: System opens a dispute log and pauses the archival step until the dispute is resolved and signed off.

Special Requirements

- System access must be revocable within one hour of termination confirmation for involuntary separations.
- All offboarding actions and approvals must be logged for legal defensibility with a full audit trail.

Technology & Data Variations	Workflow engine with task assignment and deadline tracking; integration with payroll and IT provisioning systems.
Frequency of Occurrence	Low — variable based on organizational attrition rates.
Open Issues	Exit interview survey integration and automated NOC (No Objection Certificate) generation are in the product backlog.

UC-05 | View Employee Profile

Use Case ID	UC-05
Use Case Name	View Employee Profile
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
- Employee (self-view) - HR (management view)	Admin	- Compliance Officer (audit log reviewer) - Data Privacy Officer - IT Security

Stakeholders and Interests

- Employee – wants accurate, up-to-date access to their own profile information.
- HR Officer – wants a consolidated view of all employee data for management decisions.
- Admin – wants profile access governed strictly by role-based security.
- Compliance Officer – wants complete profile history available for regulatory audits.
- Data Privacy Officer – wants all access to sensitive data logged and auditable.

Preconditions

- The requesting user is authenticated with a valid session.
- Employees may only view their own profile; HR and Admin may view any profile within their access scope.
- The target employee record exists and is in a viewable status (Active, On Leave, or Offboarding).

Success Guarantee (Postconditions)

- The requesting user is presented with employee profile data permitted under their role.
- A read-access log entry is created for any view of sensitive data fields.

Main Success Scenario

Step	Actor Action	System Response
1	User navigates to Employee Directory or uses the global search function.	System returns matching records based on the search query and the user role-based access scope.
2	User selects the target employee record.	System retrieves and renders the profile with sections: personal info, employment details, department, leave balance, attendance summary, and performance history.
3	User applies section filters to focus on a specific data category.	System dynamically renders the selected section and masks all fields outside the user role permissions.

Extensions (Alternative Flows)

- 2a. Employee record not found: System displays a Record Not Found message with a prompt to refine the search criteria.
- 2b. Employee record is archived (former employee): System displays a read-only archived profile with a clear Archived status banner.
- 3a. User attempts to access a field outside their role permissions (e.g., salary viewed by non-HR): System shows Access Restricted and creates an unauthorized access attempt log entry.
- 3b. User requests to export the employee profile to PDF: System generates a masked PDF containing only fields permitted for the requesting role.
- 3c. The profile data is temporarily unavailable due to a backend query timeout: System displays a Loading Error message with an option to retry, without exposing backend error details.
- 3d. The employee has a data privacy restriction flag on their record: System notifies the Data Privacy Officer of the access attempt and displays limited information only.

Special Requirements

- Sensitive fields including salary, national ID, and medical notes must be masked by default and require an explicit elevated permission to reveal.
- Profile page must load within 2 seconds for standard records under normal system load.

Technology & Data Variations	Web and mobile-responsive interface; role-based field masking enforced server-side.
Frequency of Occurrence	Very High — multiple accesses per day across the organization.
Open Issues	Self-service editing of minor personal details (address, emergency contact) by employees is under evaluation for a future release.

3.2 Leave and Attendance Management Module (UC-06 – UC-10)

UC-06 | Submit Leave Request

Use Case ID	UC-06
Use Case Name	Submit Leave Request
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
Employee	- HR Officer - System	- Admin - Payroll Module - Department Manager

Stakeholders and Interests

- Employee – wants leave requests submitted quickly with immediate balance feedback.
- HR Officer – wants all requests validated against policy before reaching the approval queue.
- Admin – wants organizational leave utilization tracked for workforce planning.
- Payroll Module – wants approved leave types flagged for deduction or allowance processing.
- Department Manager – wants visibility of upcoming absences for team coverage planning.

Preconditions

- The employee is authenticated and has an active employment record.
- The employee has a sufficient leave balance for the requested leave type.
- The requested dates fall within the current active leave policy calendar.

Success Guarantee (Postconditions)

- The leave request is stored in the system with status Pending Approval.
- An in-system and email notification is dispatched to the assigned HR Officer.
- The requested dates are tentatively blocked in the attendance calendar.

Main Success Scenario

Step	Actor Action	System Response
1	Employee navigates to Leave Management and clicks New Leave Request.	System displays the leave request form showing current leave balances by type.
2	Employee selects leave type, specifies start and end dates, and optionally adds a reason.	System calculates working days requested, validates balance sufficiency, and checks for blackout-period conflicts.
3	Employee submits the request.	System records the request as Pending Approval, blocks the dates in the attendance calendar, and dispatches notifications to HR.

Extensions (Alternative Flows)

- 2a. Insufficient leave balance for the selected type: System displays the remaining balance and prevents submission with a clear explanation.
- 2b. Selected dates overlap with a public holiday or organizational blackout period: System highlights the conflict and suggests adjusted dates.
- 2c. Employee already has an approved leave overlapping the requested period: System displays the conflict with the existing approved request and blocks double-booking.
- 3a. System notification service is temporarily unavailable: Request is saved successfully and the notification is queued for automatic retry.
- 3b. The requested leave type requires a supporting document (e.g., medical certificate for sick leave): System prompts the employee to upload the document and holds the request at Pending Document until uploaded.
- 3c. Employee is currently in a probation period and the leave type is restricted during probation: System blocks submission and displays the applicable policy restriction.

Special Requirements

- Leave balance must reflect all approved and pending requests in real time to prevent over-application.
- Form submission response time must not exceed 3 seconds.

Technology & Data Variations	Web and mobile interface; real-time leave balance API; email notification via SMTP; company calendar API integration.
Frequency of Occurrence	High — daily submissions, especially at month start and before public holidays.
Open Issues	Manager-level leave delegation and multi-level approval workflows are planned for a future release.

UC-07 | Approve or Reject Leave Request

Use Case ID	UC-07
Use Case Name	Approve or Reject Leave Request
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	- System - Department Manager (coverage consultant)	- Employee - Payroll Module - Admin

Stakeholders and Interests

- HR Officer – wants a clear approval queue with all necessary context to decide quickly and fairly.
- Employee – wants a timely decision with a clear reason if the request is rejected.
- Admin – wants leave approvals aligned with workforce availability thresholds.
- Payroll Module – wants approved leave types forwarded for payroll computation.
- Department Manager – wants team absences confirmed or declined before scheduling.

Preconditions

- A leave request with status Pending Approval exists in the system.
- The HR Officer is authenticated and authorized to approve leave for the requesting employee.
- The HR Officer has reviewed the team coverage calendar for the requested period.

Success Guarantee (Postconditions)

- Approved: Leave balance is deducted, attendance calendar confirmed, and employee notified.
- Rejected: Request status set to Rejected, reason recorded, and employee notified.
- A decision audit log is maintained for all approvals and rejections.

Main Success Scenario

Step	Actor Action	System Response
1	HR Officer navigates to the Pending Leave Requests queue.	System displays requests sorted by submission date with employee name, leave type, dates, and current balance.
2	HR Officer selects a request and reviews team coverage details.	System shows the full request alongside the team attendance calendar for the requested period.
3	HR Officer selects Approve or Reject and enters a mandatory comment.	System validates that a decision comment of at least 10 characters has been provided.
4	HR Officer confirms the decision.	System updates request status, adjusts balance if approved, confirms attendance calendar, and dispatches email and in-system notification to the employee.

Extensions (Alternative Flows)

- 3a. HR Officer attempts to approve when team coverage falls below the minimum threshold: System displays a coverage warning; HR must acknowledge before the approval can proceed.
- 3b. Leave balance has changed since the request was submitted (e.g., another approval reduced it to zero): System recalculates available balance before finalizing approval and alerts HR of the change.
- 3c. HR Officer approves a partial duration (e.g., approves only 3 of the 5 requested days): System records the partial approval, adjusts the balance accordingly, and prompts employee to resubmit for remaining days.
- 4a. Email service is unavailable during decision confirmation: Decision is saved and the notification is queued for automatic retry; HR is informed of the queue status.
- 4b. HR Officer attempts to approve a request for an employee who has since been offboarded: System blocks the action and displays the employee current status with an advisory to close the request.
- 4c. Leave request requires secondary Admin approval due to duration exceeding the policy maximum: System places the approval in a Secondary Review state and routes it to Admin automatically.

Special Requirements

- Decision comment must not be empty; minimum 10 characters required to ensure meaningful feedback.
- Leave balance deduction must be atomic to prevent race conditions in concurrent bulk approvals.

Technology & Data Variations	Web interface with team calendar view integration; transactional database operations for atomic balance deduction.
Frequency of Occurrence	High — multiple approval decisions expected daily.
Open Issues	Automatic escalation workflow for requests pending more than 48 hours is under development.

UC-08 | Record Daily Attendance

Use Case ID	UC-08
Use Case Name	Record Daily Attendance
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
Employee	HR Officer	- Payroll Module - Admin - IT

Stakeholders and Interests

- HR Officer – wants accurate attendance data without manual errors or manipulation.
- Employee – wants attendance correctly recorded to protect payroll and leave entitlements.
- Admin – wants automated feeds from biometric or digital systems.
- Payroll Module – wants finalized daily attendance for salary computation before cutoff.
- IT Department – wants reliable device uptime and NTP synchronization.

Preconditions

- The employee has an active record in the system.
- The current date is a designated workday in the company calendar.
- At least one active attendance capture method is available (biometric, web, or mobile).

Success Guarantee (Postconditions)

- A daily attendance record is created with check-in time, check-out time, and computed work hours.
- Late arrivals and early departures are automatically flagged for HR review.
- Payroll module is notified of attendance status for all salary-relevant days.

Main Success Scenario

Step	Actor Action	System Response
1	Employee checks in via biometric device, web portal, or mobile app.	System captures the check-in timestamp, verifies identity, and creates the daily attendance record.
2	Employee works throughout the day.	System maintains the open attendance session and monitors for extended break policy violations.
3	Employee checks out at end of the working day.	System records check-out timestamp, computes total work hours versus scheduled shift, and marks the record as complete.
4	HR Officer reviews the daily attendance log for exceptions.	System presents flagged exceptions (absent, late, early departure) for HR action or annotation.

Extensions (Alternative Flows)

- 1a. Employee forgets to check in: Attendance record is created with a Missing Punch flag; an automated alert is sent to HR and the employee.
- 1b. Biometric device is offline at check-in: System falls back to web portal or mobile app; an IT alert is triggered for device maintenance.
- 3a. Check-out not recorded by the end-of-day threshold: System auto-marks the record as Missing Check-out and dispatches an alert to HR.
- 3b. Employee checks out before the minimum shift threshold (potential early departure policy violation): System flags the record and requires HR to confirm whether the early departure was authorized.
- 4a. HR Officer identifies a system-caused error in the attendance record (e.g., due to a device sync failure): HR can apply a direct correction with a mandatory justification note, bypassing the employee correction request workflow.
- 4b. No attendance data received for an entire shift due to an integration failure: System marks all affected employees as Missing Data and raises a Priority IT Alert.

Special Requirements

- All timestamps must be synchronized with a centralized NTP time server to prevent manipulation.

- The system must support concurrent check-ins from multiple entry points without data inconsistency.

Technology & Data Variations	Biometric device integration (fingerprint / facial recognition), web portal, and mobile app; NTP-synchronized timestamps; REST API for device communication.
Frequency of Occurrence	Very High — every working day for all active employees.
Open Issues	GPS-based remote check-in for field employees is being evaluated for the next release.

UC-09 | Request Attendance Correction

Use Case ID	UC-09
Use Case Name	Request Attendance Correction
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
Employee	• HR Officer (reviewer and approver) • System	• Payroll Module • Compliance Officer • IT

Stakeholders and Interests

- Employee – wants incorrect records corrected promptly to avoid payroll and leave impacts.
- HR Officer – wants requests reviewed with clear supporting justification before approval.
- Payroll Module – wants corrections finalized before payroll processing cutoff.
- Compliance Officer – wants an immutable audit trail of all corrections with dual authorization.
- IT Department – wants system-caused errors documented separately from employee-caused errors.

Preconditions

- The employee is authenticated and the erroneous attendance record exists in the system.
- The correction request is submitted within the allowed correction window (default: 7 calendar days from the attendance date).
- The employee has not exceeded the monthly correction request limit.

Success Guarantee (Postconditions)

- Approved: Attendance record is updated with the corrected value and annotated with the correction reference ID.
- Rejected: Original record remains unchanged; rejection reason is recorded and visible to the employee.
- An audit log entry is created for the correction request and its outcome.

Main Success Scenario

Step	Actor Action	System Response
1	Employee navigates to Attendance History and identifies the erroneous record.	System displays a detailed attendance log with all timestamps, computed hours, and any existing flags.
2	Employee clicks Request Correction and fills in the form: date, field to correct, correct value, and reason.	System validates the correction window, checks the employee correction quota, and verifies logical consistency of the corrected values.
3	Employee submits the correction request.	System stores the request as Pending HR Review and notifies the HR Officer.
4	HR Officer reviews the request and approves or rejects with a mandatory comment.	System applies approved corrections to the attendance record, annotates the original entry with the correction reference, and notifies the employee of the outcome.

Extensions (Alternative Flows)

- 2a. Correction window has expired: System rejects the submission with an advisory that the employee must contact HR directly for manual correction.
- 2b. Corrected value creates a logical inconsistency (e.g., check-out recorded before check-in): System displays a validation error and prevents submission.
- 2c. Employee has reached the monthly correction request limit: System blocks the submission and displays the remaining quota and reset date.
- 4a. HR Officer identifies the error was caused by a device malfunction: HR flags the correction as System-Caused, which waives the employee quota count and bypasses the standard correction window.

- 4b. Multiple employees report identical missing punches for the same time slot (systemic issue): System groups the corrections under a Batch Correction request routed directly to IT and HR Admin.
- 4c. HR rejects the request but the employee disputes the rejection: System opens a formal dispute log and escalates to the HR Manager for final adjudication.

Special Requirements

- Original attendance values must be preserved in the audit trail; corrections must never overwrite original data.
- A maximum of two correction requests per employee per month is enforced by the system.

Technology & Data Variations	Web interface; append-only audit log table storing before and after snapshots; quota enforcement via database constraint.
Frequency of Occurrence	Low to medium — occasional, typically due to system errors or missed punches.
Open Issues	Automated correction eligibility based on documented system downtime events is being considered.

UC-10 | View Leave Balance and Attendance Summary

Use Case ID	UC-10
Use Case Name	View Leave Balance and Attendance Summary
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
• Employee (self-view) • HR Officer (team view)	• Admin • System	• Payroll Module (data consumer) • Compliance Officer • Finance

Stakeholders and Interests

- Employee – wants instant, accurate visibility of remaining leave entitlements and personal attendance record.
- HR Officer – wants consolidated team data for workforce planning and exception management.
- Admin – wants organization-level summaries for headcount and absence management.
- Payroll Module – wants verified totals before payroll finalization.
- Finance – wants absence cost data for budget planning purposes.

Preconditions

- The requesting user is authenticated.
- Leave and attendance records for the target period exist in the system.

Success Guarantee (Postconditions)

- The user is presented with accurate, real-time leave balance and attendance summary data.
- A data access event is logged for compliance purposes.

Main Success Scenario

Step	Actor Action	System Response
1	User navigates to the Leave and Attendance Dashboard.	System displays a summary widget with remaining leave balances by type and the current month attendance KPIs.
2	User selects an employee (HR) or views own data (Employee) and applies a date range filter.	System aggregates data for the selected period and renders a detailed breakdown table and trend charts.
3	User exports the summary report if required.	System generates the report in the selected format (PDF or Excel), timestamps the export, and initiates a download.

Extensions (Alternative Flows)

- 2a. No records found for the selected period: System displays a No Data Available message with a prompt to adjust the filter.
- 2b. HR Officer attempts to view summary for an employee outside their assigned scope: System blocks access and logs the unauthorized access attempt.
- 2c. Employee leave balance shows a discrepancy relative to approved requests: System flags the inconsistency with a Data Reconciliation Warning and advises contacting HR.
- 3a. Export service is temporarily unavailable: System queues the export as a background job and notifies the user by email when the file is ready.

- 3b. The selected date range exceeds 24 months: System warns of potential performance impact and offers a compressed summary view for large ranges.
- 3c. User attempts to export another employee data without permission: System blocks the export, generates the permitted self-data export only, and logs the permission override attempt.

Special Requirements

- Leave balance must reflect all approved and pending requests in real time.
- Dashboard must load within 3 seconds even with 12 months of historical data.

Technology & Data Variations	Web interface with interactive charts using Chart.js; PDF and Excel export engine; data aggregation via optimized SQL views.
Frequency of Occurrence	High — multiple accesses daily by employees and HR.
Open Issues	Comparative team-level leave heatmap for capacity planning is planned for the analytics module.

3.3 Performance and Compliance Management Module (UC-11 – UC-15)

UC-11 | Initiate Performance Evaluation Cycle

Use Case ID	UC-11
Use Case Name	Initiate Performance Evaluation Cycle
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
Admin	- HR Officer - System	- Employees - Department Managers

Stakeholders and Interests

- Admin – wants evaluation cycles configured precisely to match organizational review schedules.
- HR Officer – wants to monitor evaluation progress and issue reminders as deadlines approach.
- Employees – want fair, timely, and transparent evaluations within a defined window.
- Department Managers – want evaluation criteria aligned with departmental KPIs.
- Legal / Compliance – wants evaluation policies applied consistently.

Preconditions

- The Admin is authenticated with Performance Management configuration rights.
- No overlapping evaluation cycle is active for the selected employee groups.
- Evaluation templates and KPI frameworks are fully configured in the system.

Success Guarantee (Postconditions)

- A new evaluation cycle is created with defined scope, period, evaluators, and deadlines.
- Evaluation tasks are distributed to all designated reviewers.

- The evaluation module is opened for submissions within the configured window.
- Automated reminder notifications are scheduled for 7 days and 2 days before the deadline.

Main Success Scenario

Step	Actor Action	System Response
1	Admin navigates to Performance Management and clicks Create New Evaluation Cycle.	System displays the cycle configuration form.
2	Admin enters cycle name, evaluation period, applicable departments or employee groups, and evaluation type.	System validates the period does not conflict with any existing active cycle.
3	Admin assigns evaluators to departments or individual employees and sets the submission deadline.	System validates assignments and confirms all employees in scope have at least one evaluator.
4	Admin activates the evaluation cycle.	System creates evaluation tasks, dispatches distribution notifications, opens the evaluation module, and schedules automated reminders.

Extensions (Alternative Flows)

- 2a. Evaluation period overlaps with an active cycle: System blocks creation and displays the conflicting cycle details with the option to view or modify it.
- 3a. One or more employees in scope have no evaluator assigned: System highlights coverage gaps and prevents activation until all gaps are resolved.
- 3b. An assigned evaluator is no longer active in the system (deactivated account): System flags the inactive evaluator and requires reassignment before activation.
- 4a. Admin accidentally activates the cycle before all configuration is complete: System requires Admin to confirm activation with a checklist summary, making accidental activation harder.
- 4b. Reminder scheduling service is temporarily unavailable: Cycle is activated successfully; reminders are queued and will be sent when the scheduler recovers.
- 4c. An evaluator rejects their assignment after the cycle is active: System notifies HR and the Admin; the assignment is escalated for re-assignment with a reassignment deadline.

Special Requirements

- Evaluation cycle configurations must be locked from editing once activated; amendments require Admin-level deactivation and reactivation.
- Automated reminders must be dispatched at exactly 7 days and 2 days before the submission deadline.

Technology & Data Variations	Web interface; background task scheduling engine; email notification service with delivery confirmation.
Frequency of Occurrence	Low — typically twice per year for mid-year and annual reviews.
Open Issues	360-degree peer review integration and employee self-assessment capability are in the product roadmap.

UC-12 | Submit Employee Performance Evaluation

Use Case ID	UC-12
Use Case Name	Submit Employee Performance Evaluation
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	- Admin - System validator	- Employee - Management - Compensation Module

Stakeholders and Interests

- HR Officer / Evaluator – wants a structured, unambiguous form with clear scoring guidelines.
- Employee – wants evaluations conducted fairly and consistently with documented feedback.
- Admin – wants submission progress tracked for cycle completion monitoring.
- Management – wants aggregate data for promotion and succession planning.
- Compensation Module – wants final scores to trigger salary review workflows.

Preconditions

- An active evaluation cycle exists and the current date is within the submission window.
- The evaluator is assigned to evaluate the specific employee and is authenticated.

- The evaluation template with competencies and KPIs is fully configured.

Success Guarantee (Postconditions)

- The completed evaluation is stored and linked to the employee performance history.
- An aggregate weighted score is computed and recorded.
- The cycle completion counter is incremented.
- The employee is notified that their evaluation has been submitted.

Main Success Scenario

Step	Actor Action	System Response
1	Evaluator navigates to My Pending Evaluations and selects the employee to evaluate.	System displays the structured evaluation form pre-loaded with competencies and KPIs from the active cycle template.
2	Evaluator scores each competency and KPI on the rating scale and provides qualitative written feedback.	System calculates a running weighted score and validates that all mandatory sections are completed before final submission.
3	Evaluator reviews the evaluation summary and submits.	System validates form completeness, saves the evaluation, computes the aggregate score, marks the task as complete, and notifies the employee.

Extensions (Alternative Flows)

- 2a. Evaluator attempts to submit with one or more mandatory sections incomplete: System highlights the incomplete sections and blocks submission until all are filled.
- 3a. Submission is attempted after the deadline: System rejects the submission and displays an Overdue message; the evaluator must request a grace period extension from Admin.
- 3b. Admin grants a grace period: System re-opens the evaluation form for the specific evaluator for the duration of the grace period only.
- 3c. An evaluator identifies a conflict of interest with the employee being evaluated: Evaluator flags the conflict via the system; Admin reassigns the evaluation to a neutral party.
- 4a. System detects a statistical outlier in scores relative to the department average (calibration flag): System displays a soft warning to the evaluator suggesting a review of the scores before final submission.
- 4b. Network failure occurs during submission: System auto-saves draft data locally and prompts the evaluator to retry submission when connectivity is restored.

Special Requirements

- Submitted evaluations must be immutable; amendments require a separate correction workflow with Admin approval.

- Scoring calibration guidelines must be accessible as inline tooltips during evaluation.

Technology & Data Variations	Web interface with dynamic form rendering based on the cycle template; auto-save feature using browser local storage as a fallback.
Frequency of Occurrence	Low to medium — concentrated within active evaluation windows of 2 to 4 weeks.
Open Issues	Calibration sessions for group score fairness review are planned as a future feature.

UC-13 | Monitor Employee Probation Status

Use Case ID	UC-13
Use Case Name	Monitor Employee Probation Status
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
HR	• System • Direct Manager	• Employee • Payroll Module

Stakeholders and Interests

- Admin – wants a real-time overview of all employees on probation with clear milestone alerts.
- HR Officer – wants to record confirmation, extension, or termination decisions before probation end dates.
- Employee – wants timely and fair communication of probation outcomes.
- Legal / Compliance – wants documented, signed probation decisions for contract compliance.
- Payroll Module – wants employment type updated to Permanent upon confirmation.

Preconditions

- One or more employees have an active probation record in the system.

- The reviewing user is authenticated with Probation Management access rights.
- Interim performance evaluations for the probation period are recorded.

Success Guarantee (Postconditions)

- Probation status is updated to Confirmed, Extended, or Terminated based on the decision.
- The decision is recorded with date, decision maker ID, and supporting notes.
- Employee is notified of the probation outcome.
- If confirmed, employment type is updated to Permanent and the probation record is closed.

Main Success Scenario

Step	Actor Action	System Response
1	HR Officer navigates to the Probation Monitoring dashboard.	System displays all active probation cases with employee name, start date, end date, interim scores, and days remaining.
2	HR Officer selects an employee approaching probation completion.	System shows the full probation record including interim evaluations, manager notes, and attendance summary for the probation period.
3	HR Officer selects a decision: Confirm, Extend (with revised end date), or Terminate.	System validates the decision: extension requires a new end date; termination requires a documented reason.
4	HR Officer submits the probation decision.	System records the decision, updates employment status, notifies the employee, and archives the probation record if closed.

Extensions (Alternative Flows)

- 1a. No active probation cases exist: Dashboard displays an empty state message with a summary of recently closed probation records.
- 2a. Interim performance evaluation has not been submitted for the employee: System displays a Missing Evaluation warning and advises HR to complete the evaluation before making a decision.
- 3a. Extension selected without a revised end date: System blocks submission and requires the new probation end date.
- 3b. Total probation duration (including extensions) exceeds the maximum allowed by policy: System flags the policy limit breach and requires escalation to Admin for override authorization.
- 4a. Employee is unreachable to acknowledge the probation decision: System logs the delivery attempt and records the decision as Communicated – Unacknowledged after the defined acknowledgment window expires.

- 4b. Payroll module fails to receive the employment type update: System queues the update for retry and alerts the Payroll team manually to verify the change.

Special Requirements

- Automated alerts must be triggered at 30, 14, and 7 days before each probation end date.
- All probation decisions must be electronically acknowledged by the decision maker before becoming final.

Technology & Data Variations	Web dashboard with timeline visualization; automated email scheduler; integration with payroll for employment type update.
Frequency of Occurrence	Low to medium — depends on hiring volume; typically reviewed monthly.
Open Issues	Integration with digital signature service for formal probation confirmation letters is planned.

UC-14 | Generate Compliance Report

Use Case ID	UC-14
Use Case Name	Generate Compliance Report
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
Admin	• System • Compliance Officer (report requester and reviewer)	• Legal Department • External Auditors • IT Security

Stakeholders and Interests

- Admin – wants accurate, formatted compliance reports generated on demand and on schedule.

- Compliance Officer – wants reports meeting regulatory formatting and content requirements.
- Legal Department – wants a complete and auditable compliance activity record.
- External Auditors – want reports clearly presenting policy adherence and exception logs.
- IT Security – wants access log data included in relevant compliance domains.

Preconditions

- The Admin is authenticated with Compliance Reporting access rights.
- Relevant compliance data exists in the system for the selected reporting period.
- Report templates for the selected compliance domain are configured.

Success Guarantee (Postconditions)

- A compliance report is generated in the selected format and made available for download.
- The report is archived with a timestamp, generator ID, and version number.
- A report generation log entry is created for audit trail purposes.

Main Success Scenario

Step	Actor Action	System Response
1	Admin navigates to Reports and selects Generate Compliance Report.	System displays the compliance report configuration form with domain options and date range selectors.
2	Admin selects the compliance domain, reporting period, and output format (PDF or Excel).	System validates that data exists for the selected parameters and estimates generation time.
3	Admin initiates generation.	System aggregates compliance data, applies policy filters, formats the output, and generates the report.
4	Admin reviews the summary and downloads or schedules distribution.	System archives the report, logs the generation event, and executes scheduled distribution actions.

Extensions (Alternative Flows)

- 2a. No compliance data found for the selected domain and period: System displays a No Data Available warning and suggests broadening the date range or checking data completeness.
- 3a. Report generation exceeds 30 seconds: System queues it as a background job and notifies Admin by email when complete.
- 3b. Report generation reveals a data integrity issue in the underlying compliance records (e.g., missing mandatory field values): System flags the anomaly in the report with a Data Quality Warning annotation.

- 4a. Distribution email addresses in the scheduled delivery list are invalid: System sends the report to valid addresses, logs the failed deliveries, and alerts Admin to correct the distribution list.
- 4b. Admin attempts to delete an archived compliance report before the 5-year retention period expires: System blocks deletion with a retention policy notice and logs the deletion attempt.
- 4c. Generated report exceeds the maximum attachment size for email distribution: System uploads the report to secure storage and shares a time-limited download link instead of attaching it.

Special Requirements

- Reports must include a digital timestamp and the generating user ID on every page.
- Archived reports must be retained for a minimum of 5 years per organizational data retention policy.

Technology & Data Variations	Web interface; report generation engine (JasperReports or custom PDF/Excel builder); secure archive storage with retention policy enforcement.
Frequency of Occurrence	Low to medium — monthly, quarterly, and on-demand for audits.
Open Issues	Scheduled auto-generation and distribution of monthly compliance summaries to designated groups is in development.

UC-15 | Generate HR Analytics Report

Use Case ID	UC-15
Use Case Name	Generate HR Analytics Report
Scope	Smart HR Operations and Workforce Management System
Level	User Goal

Actors

Primary Actor	Supporting Actors	Offstage Actors
Admin	- System - HR Officer	- Management - Finance - Strategic Planning Team

Stakeholders and Interests

- Admin – wants comprehensive, data-driven insights to support strategic workforce decisions.
- HR Officer – wants analytics to identify trends in absenteeism, turnover, and performance.
- Management / C-Suite – wants high-level workforce KPIs for executive decisions.
- Finance – wants headcount and salary distribution analytics for budget planning.
- Strategic Planning Team – wants headcount forecasting data for capacity planning.

Preconditions

- The Admin is authenticated with HR Analytics access rights.
- At least one complete month of historical records exists in the system.
- The analytics module data refresh has completed within the last 24 hours.

Success Guarantee (Postconditions)

- An HR analytics report is generated with requested metrics, charts, and trend analysis.
- The report is archived in the analytics library and available for future reference.
- Export and distribution actions are completed as configured.

Main Success Scenario

Step	Actor Action	System Response
1	Admin navigates to HR Analytics and selects Generate Analytics Report.	System displays the report builder with available metric categories: headcount, attrition, leave, attendance, and performance.
2	Admin selects metrics, date range, organizational scope, and visualization type.	System validates selections and renders a preview layout of the report structure.
3	Admin initiates report generation.	System queries all HR modules, applies statistical aggregation (averages, trends, percentage changes), and renders the report with interactive charts and data tables.
4	Admin reviews the output and exports or schedules delivery.	System packages the report in the selected format, archives it, logs the event, and processes scheduled delivery.

Extensions (Alternative Flows)

- 2a. Insufficient data for a selected metric (less than one complete month): System disables the metric with an explanatory tooltip and suggests available alternatives.
- 3a. Report generation exceeds the threshold time due to large data volume: System switches to asynchronous processing and notifies Admin when the report is ready.
- 3b. Statistical outliers detected in the analytics data (e.g., spike in attrition for a single month): System annotates the chart with an Anomaly Detected flag and links to the underlying records.
- 4a. Scheduled delivery recipients include external email addresses outside the organization domain: System requires Admin confirmation before sending to external parties to comply with data privacy policy.
- 4b. The analytics data for a specific module is stale (last refresh more than 24 hours ago): System displays a Data Freshness Warning on the affected charts and offers to trigger a manual data refresh.
- 4c. Admin requests a comparison report between two organizational units that have different KPI frameworks: System aligns comparable metrics and flags non-comparable ones with a Methodology Note.

Special Requirements

- Analytics data must be refreshed from source modules at least every 24 hours.
- Interactive charts must include data labels and drill-down capability to underlying records.
- Exported reports must be formatted to executive presentation quality standards.

Technology & Data Variations	Web interface with Chart.js or embedded Power BI; asynchronous report engine; cloud storage for archive; REST API data aggregation layer.
Frequency of Occurrence	Low to medium — weekly or monthly for management reviews; on-demand for planning sessions.
Open Issues	Predictive analytics features including turnover risk modeling and leave demand forecasting are planned for the advanced analytics roadmap.

4. Domain Model Summary

The entities listed below are derived directly from the system's use case specifications and collectively represent the core operational data and business concepts required for the system to function. The concepts will be used to create the domain model which will guide the architectural design process.

4.1 Core Entities and Key Attributes

Entity	Key Attributes
Employee	employeeID, fullName, email, phoneNumber, designation, departmentID, hireDate, employmentType, employmentStatus
Department	departmentID, departmentName, managerID, parentDepartmentID
Designation	designationID, title, description
EmployeeRecord	recordID, employeeID, createdDate, lastUpdated
LeaveRequest	leaveID, employeeID, leaveType, startDate, endDate, reason, status, approverID, approvalDate
AttendanceRecord	attendanceID, employeeID, attendanceDate, checkInTime, checkOutTime, totalHours, attendanceStatus
AttendanceCorrectionRequest	correctionID, attendanceID, employeeID, originalValue, correctedValue, justification, status
PerformanceEvaluation	evaluationID, employeeID, evaluatorID, cycleID, evaluationDate, aggregateScore, remarks
EvaluationCycle	cycleID, cycleName, startDate, endDate, evaluationType, status
ProbationRecord	probationID, employeeID, startDate, endDate, decision, decisionDate, notes
ComplianceReport	reportID, reportType, generatedAt, parameters, format

Entity	Key Attributes
HRAnalyticsReport	reportID, reportPeriod, generatedAt, summaryMetrics
UserAccount	userID, employeeID, username, role, accountStatus, lastLogin

4.2 Key Relationships

The following relationships describe how the core entities interact within the HR domain.

- **Employee (1) to (*) LeaveRequest**

An employee may submit multiple leave requests throughout their employment lifecycle.

- **Employee (1) to (*) AttendanceRecord**

Each employee generates attendance records on working days, recording check-in and check-out activity.

- **Employee (1) to (*) AttendanceCorrectionRequest**

If attendance data is inaccurate, an employee may submit correction requests for review and approval.

- **Employee (1) to (*) PerformanceEvaluation**

An employee may receive performance evaluations across different evaluation cycles during their tenure.

- **Department (1) to (*) Employee**

A department contains multiple employees who belong to that organizational unit.

- **Designation (1) to (*) Employee**

Employees are assigned a designation that defines their role or position within the organization.

- **EvaluationCycle (1) to (*) PerformanceEvaluation**

A performance evaluation cycle consists of multiple employee evaluations conducted during the defined evaluation period.

- **Employee (1) to (0..1) ProbationRecord**

Newly hired employees may have an associated probation record that tracks their probation period and final confirmation decision.

- **HR reviews LeaveRequest and AttendanceCorrectionRequest**

HR personnel are responsible for validating, approving, or rejecting these requests according to organizational policies.

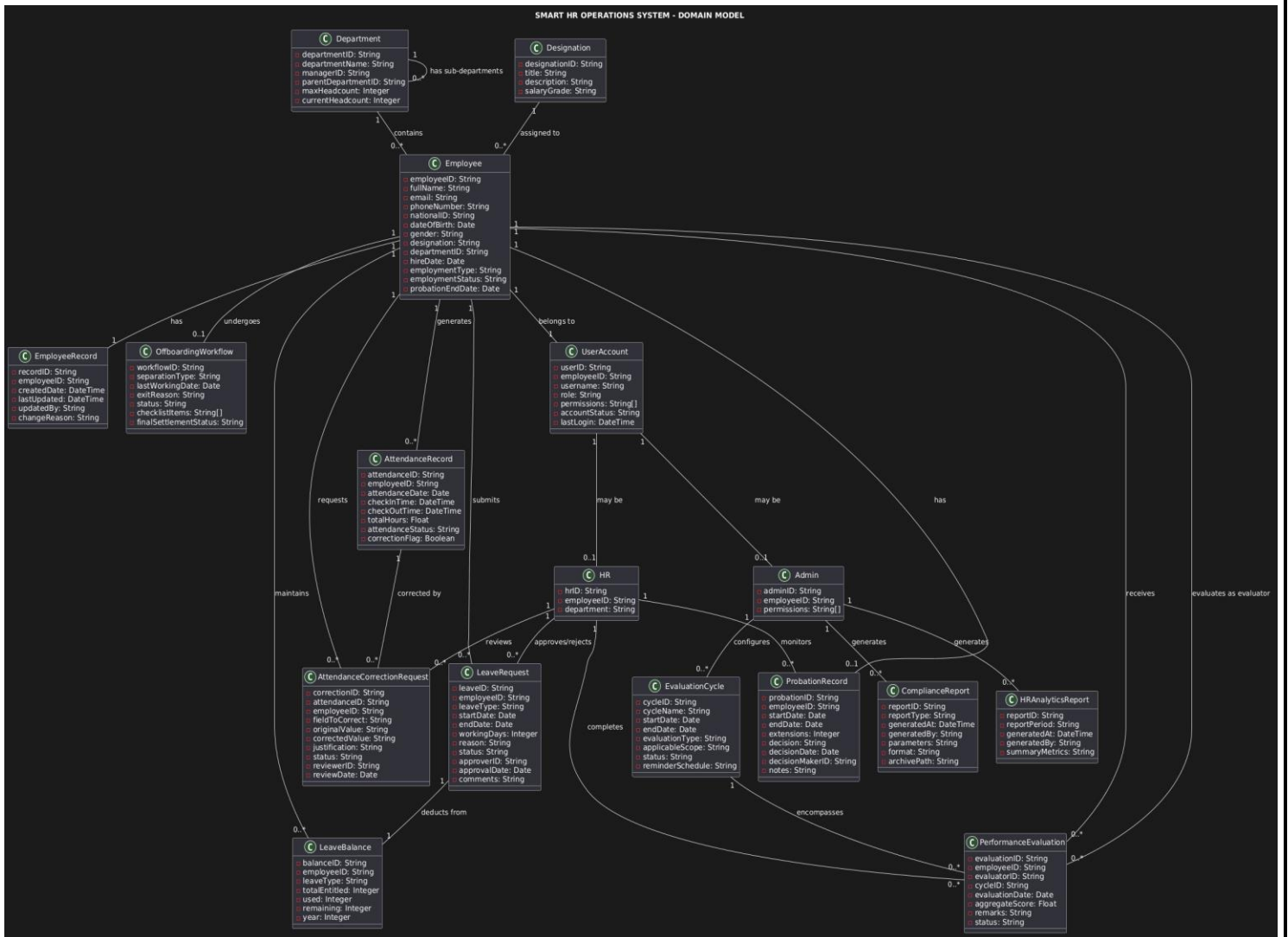
- **Admin configures EvaluationCycle and generates organizational reports**

Administrative users manage performance evaluation cycles and generate compliance or analytical reports.

- **UserAccount (1) to (1) Employee**

Each employee is associated with exactly one user account that enables secure access to the HR system through role-based permissions.

4.3 Domain Model



5. Use Case Diagram

